



Two-dimensional quaternion linear canonical series for color images

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ABSTRACT

The quaternion-based moments and transforms are among the essential tools for color image processing, and received increasing attention and interest from researchers. In this paper, a novel two-dimensional quaternion linear canonical series (2-D QLCS) for describing color images is introduced. The capabilities of the proposed 2-D QLCS on image representation and reconstruction are analyzed. The proposed 2-D QLCS can be acquired from the common two-dimensional linear canonical transform series (2-D LCTS). The relationships between 2-D QLCS and 2-D LCTS are investigated in detail. Experimental results verify the correctness of the proposed method and show that 2-D QLCS has ideal performance in terms of scale invariance, numerical stabilities and image representation.

1. Introduction

Nowadays, digital images play an important part in our life with the development of computer and communication technology, and a variety of image processing technologies have been proposed. Therefore, the invariant moments or transforms have drawn a large number of researchers, and widely been applied in image analysis [1–5], face recognition [6–8], image watermarking [9–11], forgery detection [12, 13] and other applications [14–16]. The methods based on moments or transforms become one of the important tools in the field of image processing.

Moments can be regarded as the projections of an image function onto specific kernel functions, in Cartesian coordinates:

$$M_{nm}^c = \int \int_{\mathbb{R}^2} h_{n,m}(x, y) f(x, y) dx dy, \quad (1)$$

or in polar coordinates:

$$M_{nm}^p = \int_0^{2\pi} \int_0^1 h_{n,m}(r, \theta) f(r, \theta) r dr d\theta, \quad (2)$$

where $f(x, y)$ and $f(r, \theta)$ denotes the image function represented in Cartesian coordinates and polar coordinates, respectively. M_{nm}^c and M_{nm}^p denotes the $(n+m)$ th-order moments of image function $f(x, y)$ and $f(r, \theta)$, respectively. $n, m = 0, 1, 2, \dots$, and $h_{n,m}(x, y)$ and $h_{n,m}(r, \theta)$ are the kernel functions (also known as the basis functions). Different types of moments can be obtained based on the different kernel functions.

In general, moments and transforms are classified as the orthogonal and the non-orthogonal type based on the orthogonality of the transform kernel functions. Non-orthogonal moments mainly consist

of geometric moments [17], rotational moments [18] and complex moments [19]. However, the non-orthogonal moments with information redundancy are unavoidable because of the non-orthogonality, and the reconstruction of images becomes much more difficult. By contrast, orthogonal moments have good performance of image reconstruction with the orthogonality and have been extensively researched. Various kinds of the orthogonal moments have been proposed and investigated in recent years, such as Zernike moments (ZM) [20], pseudo-Zernike moments (PZM) [21], orthogonal Fourier–Mellin moments (OFMM) [22], radial harmonic Fourier moments (RHFM) [23], Chebyshev–Fourier moments (CHFM) [24], Bessel–Fourier moments (BFM) [25], Jacobi–Fourier moments (JFM) [26], Exponent–Fourier moments (EFM) [27], Tchebichef moments (TM) [28], Krawtchouk moments (KM) [29], Racah moments (RM) [30], dual Hahn moments (DHM) [31], and so on.

Among the orthogonal moments, the polar harmonic transforms (PHTs) based on the trigonometric functions are first proposed by Yap et al. [32]. PHTs have better performance in image reconstruction, lower computational complexity and higher noise robustness than other orthogonal moments. Furthermore, PHTs have been widely applied in fingerprint classification [33] and image watermarking [34]. On this basis, Qi et al. [35] introduced the two-dimensional linear canonical transform series (2-D LCTS) and the polar linear canonical transform (PLCT), and have been successfully applied in image watermarking [9, 36]. Those results are mainly focus on grayscale images.

However, almost all the images acquired in our lives are chromatic as a result of the popularity of digital cameras. Moreover, the color images attracted more and more attention from researchers because

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they supply richer information than grayscale images. Nowadays, the quaternion-based moments and transforms are one of the essential tools for color image processing [3,5,13–15,37–49]. The main superiority of representing color images by quaternion is that a color image can be regarded as a vector holistically without losing color information [39]. The quaternion-based methods exploited in color image processing have been introduced by Sangwine [37], Pei and Cheng [38]. From then on, based on the theory of quaternion, many traditional methods directed at grayscale images are extended to the quaternion domain for color image processing. A large number of the quaternion-based orthogonal moments have been proposed and investigated, such as quaternion Fourier–Mellin moments (QOFMM) [40], quaternion Zernike moments (QZM) [41], quaternion pseudo-Zernike moments (QPZM) [3], quaternion polar harmonic transforms (QPHTs) [42], quaternion polar linear canonical transform (QPLCT) [43], quaternion polar harmonic Fourier moments (QPHFM) [5], quaternion Bessel–Fourier moments (QBFM) [50], quaternion exponent moments (QEM) [51], quaternion radial harmonic Fourier moments (QRHFM) [52], and so on.

As a new kind of PHTs, 2-D LCTS holds better properties than PHTs [35]. However, the 2-D LCTS fails to process color image in a holistic manner, which greatly limits its application in color image processing. In order to overcome this shortcoming, the main objective of this paper is to propose a novel method to be used in color image analysis based on the 2-D LCTS. In [44,48], Bahri et al. introduced a definition of the quaternion linear canonical transform (QLCT) and investigate some important properties. On this basis, the main contributions of this paper can be stated as follows:

First, a new quaternion orthogonal transform, named two-dimensional quaternion linear canonical series (2-D QLCS) is proposed in this paper. Second, the capabilities of the proposed 2-D QLCS on image representation and reconstruction are analyzed. Third, The relationships between 2-D QLCS and 2-D LCTS are investigated in detail. Finally, several comparison experiments of 2-D QLCS, QZM, QPHTs and QPHFM are conducted. The experimental results verify the correctness of the proposed method and show that 2-D QLCS perform better in scale invariance, numerical stabilities and image representation than the other competing methods.

This paper is organized as follows. In Section 2, a brief introduction of the quaternion algebra and 2-D LCTS is provided. In Section 3, the 2-D QLCS is defined and the related properties are discussed in detail. Experimental results and analysis are given in Section 4. Section 5 concludes this paper.

2. Preliminaries

2.1. Quaternion algebra

The quaternion number, a type of hypercomplex number, was formally introduced by Hamilton in 1843 [53]. A quaternion consists of one real part and three imaginary parts as follows:

$$q = q_r + q_i i + q_j j + q_k k, \quad (3)$$

where q_r , q_i , q_j and q_k are real numbers, and i , j and k are complex operators with following properties:

$$i^2 = j^2 = k^2 = ijk = -1, \quad (4)$$

$$ij = -ji = k, jk = -kj = i, ki = -ik = j. \quad (5)$$

For a quaternion q , the conjugate and the norm of q are respectively given by:

$$q^* = q_r - q_i i - q_j j - q_k k, \quad |q| = \sqrt{q \cdot q^*} = \sqrt{q_r^2 + q_i^2 + q_j^2 + q_k^2}. \quad (6)$$

A quaternion q can be represented as the sum of its scalar and vector parts: $q = S(q) + V(q)$, where $S(q) = q_r$, $V(q) = q_i i + q_j j + q_k k$. If a quaternion q has a zero scalar part ($q_r = 0$), then q is called pure

quaternion, and if q has a unit norm ($|q| = 1$), then q is called unit quaternion. Moreover, the quaternion q can be represented in polar form: $q = |q| e^{\rho\phi}$, where $\rho = V(q)/|V(q)|$ and $\phi = \arctan|V(q)|/S(q)$ are called eigenaxis and eigenangle, respectively.

For any two quaternions p and q , the multiplication of their conjugates satisfies: $(p \cdot q)^* = q^* \cdot p^*$.

2.2. 2-D LCTS

Let $A = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}$, $B = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$ be two matrix parameters satisfying $\det(A) = \det(B) = 1$. The two-dimensional linear canonical transform series (2-D LCTS) [35] for image $f(x, y)$ onto a domain $(x, y) \in [-1, 1] \times [-1, 1]$ is defined as follows:

$$M_{nm}^{2-D \text{ LCTS}} = \iint_D \Psi_{nm}^{A, B^*}(x, y) f(x, y) dx dy, \quad (7)$$

where

$$\Psi_{nm}^{A, B}(x, y) = \frac{i}{2} \exp \left\{ \frac{i}{2b_1} [d_1(n\pi b_1)^2 - 2x(n\pi b_1) + a_1 x^2] \right\} \cdot \exp \left\{ \frac{i}{2b_2} [d_2(m\pi b_2)^2 - 2y(m\pi b_2) + a_2 y^2] \right\}, \quad (8)$$

$D = \{(x, y) | |x|, |y| \leq 1\}$, and $*$ denotes the complex conjugate.

For image $f(x, y)$ with size $N \times M$, let $g[k, l]$, $k = 0, 1, \dots, N-1$, $l = 0, 1, \dots, M-1$ denotes the discrete form of $f(x, y)$. Project f onto a domain $(x_k, y_l) \in [-1, 1] \times [-1, 1]$ with

$$x_k = \frac{k - N/2 + 1}{N/2}, y_l = \frac{l - M/2 + 1}{M/2}. \quad (9)$$

The discrete form of 2-D LCTS in Cartesian coordinates can be represented as

$$M_{nm}^{2-D \text{ LCTS}} = \sum_{l=0}^{M-1} \sum_{k=0}^{N-1} \Psi_{nm}^{A, B^*}(x_k, y_l) f(x_k, y_l) \Delta x \Delta y, \quad (10)$$

subject to $x_k^2 + y_l^2 \leq 1$, where $\Psi_{nm}^{A, B}$ is given by Eq. (8), and $f(x_k, y_l) = g[k, l]$, $\Delta x = 2/N$, $\Delta y = 2/M$.

The image $f(x, y)$ can be approximately reconstructed by its limited 2-D LCTS coordinates as follows:

$$f(x, y) = \sum_{(n, m) \in H_K} \Psi_{nm}^{A, B}(x, y) M_{nm}^{2-D \text{ LCTS}}, \quad (11)$$

where $H_K = \{(n, m) | |n| + |m| \leq K\}$, K is a finite number.

In this paper, the 2-D LCTS is represented by M_{nm} , unless otherwise stated.

3. 2-D quaternion linear canonical series for color images

3.1. 2-D quaternion linear canonical series

In [37], based on the theory of quaternion algebraic, a color image $f(x, y)$ can be represented as a pure quaternion number:

$$f(x, y) = f_R(x, y)i + f_G(x, y)j + f_B(x, y)k, \quad (12)$$

where $f_R(x, y)$, $f_G(x, y)$, and $f_B(x, y)$ represent the red, green and blue components, respectively.

Due to the non-commutativity of quaternions multiplication, there are two different definitions [40] of 2-D QLCS, as follows:

Definition 1. Let $A = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}$, $B = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$ be two matrix parameters satisfying $\det(A) = \det(B) = 1$. For color image $f(x, y)$, the left-side 2-D quaternion linear canonical series is defined as:

$$M_{nm}^{L, 2-D \text{ QLCS}} = \iint_{D=\{(x, y) | |x|, |y| \leq 1\}} \Phi_{nm}^{A, B^*}(x, y) f(x, y) dx dy, \quad (13)$$

where

$$\Phi_{nm}^{A,B}(x, y) = \frac{\mu}{2} \exp \left\{ \frac{\mu}{2b_1} [d_1(n\pi b_1)^2 - 2x(n\pi b_1) + a_1 x^2] \right\} \cdot \exp \left\{ \frac{\mu}{2b_2} [d_2(m\pi b_2)^2 - 2y(m\pi b_2) + a_2 y^2] \right\}, \quad (14)$$

$\mu = \alpha i + \beta j + \gamma k$ is a unit pure quaternion.

Definition 2. Let $A = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}$, $B = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$ be two matrix parameters satisfying $\det(A) = \det(B) = 1$. For color image $f(x, y)$, the right-side 2-D QLCS can be defined by reversing the orders of image and transform kernel in Eq. (13):

$$M_{nm}^{R, 2-D QLCS} = \iint_{D=\{(x,y)| |x|, |y| \leq 1\}} f(x, y) \Phi_{nm}^{A,B*}(x, y) dx dy. \quad (15)$$

Similar to Eq. (11), the color image $f(x, y)$ can be approximately reconstructed by its limited 2-D QLCS coordinates, and the reconstructions based on left-side and right-side 2-D QLCS coefficients can be written as follows:

$$f(x, y) = \sum_{(n,m) \in H_K} \Phi_{nm}^{A,B}(x, y) M_{nm}^{L, 2-D QLCS}, \quad (16)$$

$$f(x, y) = \sum_{(n,m) \in H_K} M_{nm}^{R, 2-D QLCS} \Phi_{nm}^{A,B}(x, y), \quad (17)$$

where $H_K = \{(n, m) | |n| + |m| \leq K\}$, K is a finite number, and $\Phi_{nm}^{A,B}(x, y)$ is given by Eq. (14).

In addition, it is worth noting that the left-side and right-side 2-D QLCS are different transformations based on the non-commutativity of quaternions multiplication, and it is necessary to study their performance respectively. In this section, the 2-D QLCS refer to the left-side 2-D QLCS and denoted as M_{nm}^L , unless otherwise stated. The right-side 2-D QLCS can be studied by using the similarity method, and we omit it in this paper. The performance analysis of the left- and right-side 2-D QLCS are given in the experimental part of this paper. In addition, based on the kernel function shown in Eq. (14), the transform proposed in this paper is different from QLCT that proposed in Ref. [44,48].

3.2. Scaling invariance of 2-D QLCS

The scale invariance of moments means that the arbitrary $(n + m)$ -th-order moments remain unchanged under different scale transformations. The scaling invariance can be obtained by normalizing the image into a unit domain [5].

In this paper, the image $f(x, y)$ with $N \times M$ pixels is mapped to a domain $(x_k, y_l) \in [-1, 1] \times [-1, 1]$ with

$$x_k = \frac{k - N/2 + 1}{N/2}, y_l = \frac{l - M/2 + 1}{M/2}, k = 0, 1, \dots, N - 1, l = 0, 1, \dots, M - 1, \quad (18)$$

before computing 2-D QLCS coefficients, and the 2-D QLCS coefficients are inherently invariant to scaling. Therefore, it is the normalization that guarantees the invariance of 2-D QLCS against scaling.

3.3. Relationship between 2-D QLCS and 2-D LCTS on the calculation

In this subsection, we study the relationship between the calculation of 2-D QLCS and the single channel computation of complex 2-D LCTS for grayscale image. According to the definition of 2-D LCTS, for the grayscale image $f(x, y)$, we have

$$M_{nm} = \iint_{x^2+y^2 \leq 1} \frac{-i}{2} \exp \left\{ \frac{-i}{2b_1} [d_1(n\pi b_1)^2 - 2x(n\pi b_1) + a_1 x^2] \right\} \cdot \exp \left\{ \frac{-i}{2b_2} [d_2(m\pi b_2)^2 - 2y(m\pi b_2) + a_2 y^2] \right\} f(x, y) dx dy. \quad (19)$$

For convenience, we define

$$A_{n,x} = \frac{d_1(n\pi b_1)^2 - 2x(n\pi b_1) + a_1 x^2}{2b_1}, \quad (20)$$

and

$$B_{m,y} = \frac{d_2(m\pi b_2)^2 - 2y(m\pi b_2) + a_2 y^2}{2b_2}. \quad (21)$$

Thus, Eq. (19) can be rewritten as:

$$\begin{aligned} M_{nm} &= \iint_D \frac{-i}{2} \exp \{-iA_{n,x}\} \cdot \exp \{-iB_{m,y}\} f(x, y) dx dy \\ &= \iint_D \frac{-i}{2} (\cos A_{n,x} - i \sin A_{n,x}) (\cos B_{m,y} - i \sin B_{m,y}) f(x, y) dx dy \\ &= \iint_D \frac{-i}{2} [\cos A_{n,x} \cos B_{m,y} - i (\cos A_{n,x} \sin B_{m,y} + \cos B_{m,y} \sin A_{n,x}) \\ &\quad - \sin A_{n,x} \sin B_{m,y}] f(x, y) dx dy \\ &= \iint_D -\frac{\cos A_{n,x} \sin B_{m,y} + \cos B_{m,y} \sin A_{n,x}}{2} f(x, y) dx dy \\ &\quad + \iint_D \frac{-i}{2} (\cos A_{n,x} \cos B_{m,y} - \sin A_{n,x} \sin B_{m,y}) f(x, y) dx dy \\ &:= \text{Re}(M_{nm}) + \text{Im}(M_{nm})i, \end{aligned} \quad (22)$$

where $\text{Re}(M)$ and $\text{Im}(M)$ denotes the real part and the imaginary part of the complex number M , respectively.

Without loss of generality, $\mu = \alpha i + \beta j + \gamma k$ is used to calculate the 2-D QLCS. Inspired by single channel computation of complex 2-D LCTS for grayscale image, the specific computation of 2-D QLCS is as following:

$$\begin{aligned} M_{nm}^L &= \iint_D \exp \{-\mu B_{m,y}\} \cdot \exp \{-\mu A_{n,x}\} \cdot \frac{-\mu}{2} f(x, y) dx dy \\ &= \iint_D (\cos B_{m,y} - \mu \sin B_{m,y}) (\cos A_{n,x} - \mu \sin A_{n,x}) \cdot \frac{-\mu}{2} f(x, y) dx dy \\ &= \iint_D [\cos A_{n,x} \cos B_{m,y} - \mu (\cos A_{n,x} \sin B_{m,y} + \cos B_{m,y} \sin A_{n,x}) \\ &\quad - \sin A_{n,x} \sin B_{m,y}] \cdot \frac{-\mu}{2} f(x, y) dx dy \\ &= \iint_D -\frac{\cos A_{n,x} \sin B_{m,y} + \cos B_{m,y} \sin A_{n,x}}{2} f(x, y) dx dy \\ &\quad + \iint_D (\cos A_{n,x} \cos B_{m,y} - \sin A_{n,x} \sin B_{m,y}) \cdot \frac{-\mu}{2} f(x, y) dx dy \\ &= \iint_D -\frac{\cos A_{n,x} \sin B_{m,y} + \cos B_{m,y} \sin A_{n,x}}{2} (f_R(x, y)i + f_G(x, y)j \\ &\quad + f_B(x, y)k) dx dy + \iint_D (\cos A_{n,x} \cos B_{m,y} - \sin A_{n,x} \sin B_{m,y}) \\ &\quad \cdot \frac{-\mu}{2} (f_R(x, y)i + f_G(x, y)j + f_B(x, y)k) dx dy \\ &= \text{Re}(M_{nm}(f_R))i + \text{Re}(M_{nm}(f_G))j + \text{Re}(M_{nm}(f_B))k \\ &\quad + \iint_D -\frac{\cos A_{n,x} \cos B_{m,y} - \sin A_{n,x} \sin B_{m,y}}{2} \cdot (\alpha i + \beta j + \gamma k) \\ &\quad \cdot (f_R(x, y)i + f_G(x, y)j + f_B(x, y)k) dx dy \\ &= \text{Re}(M_{nm}(f_R))i + \text{Re}(M_{nm}(f_G))j + \text{Re}(M_{nm}(f_B))k \\ &\quad + \iint_D -\frac{\cos A_{n,x} \cos B_{m,y} - \sin A_{n,x} \sin B_{m,y}}{2} \cdot (-\alpha f_R(x, y) \\ &\quad - \beta f_G(x, y) - \gamma f_B(x, y) + \alpha f_G(x, y)j + \alpha f_B(x, y)ik + \beta f_R(x, y)ji \\ &\quad + \beta f_B(x, y)jk + \gamma f_R(x, y)ki + \gamma f_G(x, y)kj) dx dy \\ &:= R_{nm}^{2-D QLCS} + I_{nm}^{2-D QLCS}i + J_{nm}^{2-D QLCS}j + K_{nm}^{2-D QLCS}k, \end{aligned} \quad (23)$$

with

$$\begin{aligned} R_{nm}^{2-D QLCS} &= -\alpha \text{Im}(M_{nm}(f_R)) - \beta \text{Im}(M_{nm}(f_G)) - \gamma \text{Im}(M_{nm}(f_B)), \\ I_{nm}^{2-D QLCS} &= \text{Re}(M_{nm}(f_R)) + \beta \text{Im}(M_{nm}(f_B)) - \gamma \text{Im}(M_{nm}(f_G)), \\ J_{nm}^{2-D QLCS} &= \text{Re}(M_{nm}(f_G)) + \gamma \text{Im}(M_{nm}(f_R)) - \alpha \text{Im}(M_{nm}(f_B)), \\ K_{nm}^{2-D QLCS} &= \text{Re}(M_{nm}(f_B)) + \alpha \text{Im}(M_{nm}(f_G)) - \beta \text{Im}(M_{nm}(f_R)), \end{aligned} \quad (24)$$

where f_R , f_G , and f_B represent the red, green, and blue components of the color image $f(x, y)$, respectively; the corresponding 2-D LCTS are respectively represented by $M_{nm}(f_R)$, $M_{nm}(f_G)$, and $M_{nm}(f_B)$. Obviously, each component of the 2-D QLCS can be obtained by the complex 2-D LCTS.

3.4. Relationship between 2-D QLCS and 2-D LCTS on image reconstruction

In this subsection, we study the relationship on image reconstruction between the 2-D QLCS and the complex 2-D LCTS. Similar to the calculation of 2-D QLCS, the reconstruction of 2-D QLCS can be acquired from the complex 2-D LCTS. The detailed process is:

$$\begin{aligned} f'(x, y) &= \sum_{l=0}^{M-1} \sum_{k=0}^{N-1} \Phi_{nm}^{A,B}(x, y) M_{nm}^L \Delta x \Delta y \\ &= \sum_{l=0}^{M-1} \sum_{k=0}^{N-1} \frac{\mu}{2} \cdot \exp\{\mu A_{n,x}\} \cdot \exp\{\mu B_{m,y}\} \\ &\quad \cdot (R_{nm}^{2-D \text{ QLCS}} + I_{nm}^{2-D \text{ QLCS}} i + J_{nm}^{2-D \text{ QLCS}} j + K_{nm}^{2-D \text{ QLCS}} k) \Delta x \Delta y \\ &= \sum_{l=0}^{M-1} \sum_{k=0}^{N-1} \frac{\mu}{2} \cdot \exp\{\mu A_{n,x}\} \cdot \exp\{\mu B_{m,y}\} \cdot R_{nm}^{2-D \text{ QLCS}} \Delta x \Delta y \\ &\quad + \left[\sum_{l=0}^{M-1} \sum_{k=0}^{N-1} \frac{\mu}{2} \cdot \exp\{\mu A_{n,x}\} \cdot \exp\{\mu B_{m,y}\} \cdot I_{nm}^{2-D \text{ QLCS}} \Delta x \Delta y \right] \cdot i \\ &\quad + \left[\sum_{l=0}^{M-1} \sum_{k=0}^{N-1} \frac{\mu}{2} \cdot \exp\{\mu A_{n,x}\} \cdot \exp\{\mu B_{m,y}\} \cdot J_{nm}^{2-D \text{ QLCS}} \Delta x \Delta y \right] \cdot j \\ &\quad + \left[\sum_{l=0}^{M-1} \sum_{k=0}^{N-1} \frac{\mu}{2} \cdot \exp\{\mu A_{n,x}\} \cdot \exp\{\mu B_{m,y}\} \cdot K_{nm}^{2-D \text{ QLCS}} \Delta x \Delta y \right] \cdot k. \end{aligned}$$

We define

$$\begin{aligned} R'_{nm} &= \sum_{l=0}^{M-1} \sum_{k=0}^{N-1} \frac{i}{2} \cdot \exp\{i A_{n,x}\} \cdot \exp\{i B_{m,y}\} \cdot R_{nm}^{2-D \text{ QLCS}} \Delta x \Delta y \\ &:= \text{Re}(R'_{nm}) + i \text{Im}(R'_{nm}), \\ I'_{nm} &= \sum_{l=0}^{M-1} \sum_{k=0}^{N-1} \frac{i}{2} \cdot \exp\{i A_{n,x}\} \cdot \exp\{i B_{m,y}\} \cdot I_{nm}^{2-D \text{ QLCS}} \Delta x \Delta y \\ &:= \text{Re}(I'_{nm}) + i \text{Im}(I'_{nm}), \\ J'_{nm} &= \sum_{l=0}^{M-1} \sum_{k=0}^{N-1} \frac{i}{2} \cdot \exp\{i A_{n,x}\} \cdot \exp\{i B_{m,y}\} \cdot J_{nm}^{2-D \text{ QLCS}} \Delta x \Delta y \\ &:= \text{Re}(J'_{nm}) + i \text{Im}(J'_{nm}), \\ K'_{nm} &= \sum_{l=0}^{M-1} \sum_{k=0}^{N-1} \frac{i}{2} \cdot \exp\{i A_{n,x}\} \cdot \exp\{i B_{m,y}\} \cdot K_{nm}^{2-D \text{ QLCS}} \Delta x \Delta y \\ &:= \text{Re}(K'_{nm}) + i \text{Im}(K'_{nm}). \end{aligned} \quad (25)$$

It is obvious that the grayscale reconstruction matrices of $R_{nm}^{2-D \text{ QLCS}}$, $I_{nm}^{2-D \text{ QLCS}}$, $J_{nm}^{2-D \text{ QLCS}}$, and $K_{nm}^{2-D \text{ QLCS}}$ are respectively represented by R'_{nm} , I'_{nm} , J'_{nm} , and K'_{nm} . Based on the Eqs. (22) and (23), the Eq. (25) can be rewritten as

$$\begin{aligned} f'(x, y) &= \sum_{l=0}^{M-1} \sum_{k=0}^{N-1} \Phi_{nm}^{A,B}(x, y) M_{nm}^L \Delta x \Delta y \\ &= \text{Re}(R'_{nm}) + (\alpha i + \beta j + \gamma k) \text{Im}(R'_{nm}) \\ &\quad + [\text{Re}(I'_{nm}) + (\alpha i + \beta j + \gamma k) \text{Im}(I'_{nm})] \cdot i \\ &\quad + [\text{Re}(J'_{nm}) + (\alpha i + \beta j + \gamma k) \text{Im}(J'_{nm})] \cdot j \\ &\quad + [\text{Re}(K'_{nm}) + (\alpha i + \beta j + \gamma k) \text{Im}(K'_{nm})] \cdot k \\ &= f'_r(x, y) + f'_i(x, y)i + f'_j(x, y)j + f'_k(x, y)k, \end{aligned} \quad (26)$$

with

$$\begin{aligned} f'_r(x, y) &= \text{Re}(R'_{nm}) - \alpha \text{Im}(I'_{nm}) - \beta \text{Im}(J'_{nm}) - \gamma \text{Im}(K'_{nm}), \\ f'_i(x, y) &= \text{Re}(I'_{nm}) + \alpha \text{Im}(R'_{nm}) + \beta \text{Im}(K'_{nm}) - \gamma \text{Im}(J'_{nm}), \\ f'_j(x, y) &= \text{Re}(J'_{nm}) + \beta \text{Im}(R'_{nm}) + \gamma \text{Im}(I'_{nm}) - \alpha \text{Im}(K'_{nm}), \\ f'_k(x, y) &= \text{Re}(K'_{nm}) + \gamma \text{Im}(R'_{nm}) + \alpha \text{Im}(J'_{nm}) - \beta \text{Im}(I'_{nm}), \end{aligned} \quad (27)$$

where $\Delta x = 2/N$, $\Delta y = 2/M$, $f'_r(x, y)$ is a matrix approximate to 0; the red, green, and blue channel of the reconstructed color image $f(x, y)$ are represented by $f'_r(x, y)$, $f'_i(x, y)$, and $f'_k(x, y)$, respectively.

4. Experimental results

In this section, several experiments are carried out to test the performance of the proposed 2-D QLCS. L-2-D QLCS and R-2-D QLCS denote left- and right-side 2-D QLCS, respectively. In the following subsections, the unit pure quaternion as $\mu = (i + j + k)/\sqrt{3}$ is specified. The reconstruction performance is evaluated by the Root-Mean-Squared Error (RMSE), a smaller RMSE value indicates better reconstruction performance, RMSE is defined as follows:

$$RMSE = \sqrt{\frac{\sum_{x^2+y^2 \leq 1} [f(x, y) - \hat{f}(x, y)]^2}{|(x, y)|x^2 + y^2 \leq 1}}. \quad (28)$$

4.1. Experiment on impact of the parameter matrix A and B

In this subsection, the aim is to evaluate the impact of parameter matrix A and B on the magnitudes and the image reconstruction ability of 2-D QLCS. The parameter matrix

$$\begin{aligned} A_1 &= (0.1402, 1.1040; -0.7916, 0.8994), \\ B_1 &= (0.1542, 1.5813; -0.4916, 1.4443); \\ A_2 &= (1.7508, 1.8285; -0.0593, 0.5092), \\ B_2 &= (6.1586, 6.3452; 5.7849, 6.1225); \\ A_3 &= (2.9965, 3.1536; 1.7649, 2.1911), \\ B_3 &= (0.8698, 2.2795; -0.4088, 0.0784); \\ A_4 &= (12.7529, 29.0698; 12.0834, 27.6211), \\ B_4 &= (1.0388, 1.3063; -0.3691, 0.4985); \\ A_5 &= (0.1773, 1.2495; -0.7142, 0.6065), \\ B_5 &= (0.8651, 1.0921; -0.5508, 0.4606); \\ A_6 &= (0.3547, 1.0420; -0.7829, 0.5193), \\ B_6 &= (3.3570, 4.4683; 1.7411, 2.6153); \end{aligned}$$

are randomly selected as test matrix, and the color image Lena with 512×512 pixels in Fig. 1(a) is selected as test image. The RMSE values are calculated to measure the distortions, which can be used to evaluate the image representation abilities. The simulation results are listed in Tables 1 and 2. In Table 1, $|M_{n,m}^q|$ represents the magnitudes of 2-D QLCS of order n with repetition m for color image Lena.

As shown in Tables 1 and 2, the different parameter matrices A and B have limited influence on the magnitudes and RMSE values with the same number of order n, m and K . It can be seen that the 2-D QLCS has good performance of image representation, regardless of the choice of A and B.

4.2. Experiment on scaling invariance of 2-D QLCS

In this subsection, several experiments are carried out to evaluate the effectiveness of the 2-D QLCS invariance under image scaling. The experiment results show that the 2-D QLCS performs better in scale invariance than QPHFM, QPCET and QZM. The original color images are Lena and pepper in Fig. 1, and the scaled images of the original ones are shown in Figs. 2 and 3. The experimental results are shown in Tables 3 to 6.

According to the experiments, the proposed 2-D QLCS invariants include L- and R-side 2-D QLCS invariants remain nearly constant under different scale transformations, and the standard deviations are generally less than 0.001. Therefore, the 2-D QLCS invariants derived in this paper could be a useful tool in pattern recognition. Note that the 2-D QLCS is not invariant to image rotation and translation.

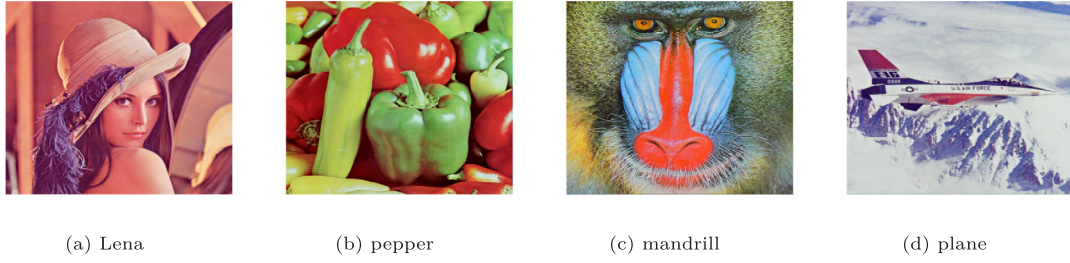
Fig. 1. The test images with 512×512 pixels.

Table 1
Magnitudes of left- and right-side 2-D QLCS for the color image Lena.

		$M_{0,0}^q$	$M_{1,0}^q$	$M_{1,1}^q$	$M_{1,2}^q$	$M_{2,1}^q$	$M_{1,3}^q$	$M_{3,1}^q$	$M_{2,2}^q$
L-2-D QLCS	A_1, B_1	1.43848	0.25510	0.20524	0.06525	0.02321	0.06544	0.04228	0.11505
	A_2, B_2	1.42559	0.25328	0.20265	0.05645	0.03803	0.07012	0.04313	0.10929
	A_3, B_3	1.43021	0.25522	0.20210	0.06197	0.03959	0.06794	0.04609	0.11095
	A_4, B_4	1.43238	0.25640	0.19888	0.05901	0.02719	0.06892	0.04085	0.11133
	A_5, B_5	1.43232	0.26459	0.19653	0.06005	0.02182	0.06861	0.03909	0.11187
	A_6, B_6	1.43323	0.25820	0.19854	0.05958	0.02539	0.06864	0.04047	0.11174
	Stand. dev.	0.00419	0.00400	0.00321	0.00298	0.00768	0.00156	0.00246	0.00188
R-2-D QLCS	A_1, B_1	1.43824	0.24797	0.21475	0.04755	0.02523	0.06992	0.04163	0.11792
	A_2, B_2	1.42349	0.24659	0.21660	0.03912	0.03965	0.07416	0.04245	0.11253
	A_3, B_3	1.42894	0.24938	0.21499	0.04449	0.04114	0.07271	0.04535	0.11401
	A_4, B_4	1.43093	0.24901	0.21094	0.04198	0.02925	0.07275	0.04018	0.11452
	A_5, B_5	1.43110	0.25687	0.20763	0.04350	0.02445	0.07220	0.03851	0.11500
	A_6, B_6	1.43191	0.25071	0.21019	0.04260	0.02758	0.07244	0.03983	0.11489
	Stand. dev.	0.00477	0.00360	0.00345	0.00280	0.00733	0.00138	0.00241	0.00177

Table 2
RMSEs of left- and right-side 2-D QLCS for the color image Lena.

		$K = 5$	$K = 10$	$K = 40$	$K = 91$	$K = 121$	$K = 199$	$K = 290$	$K = 317$
L-2-D QLCS	A_1, B_1	0.130211	0.0797	0.033126	0.017739	0.013486	0.007636	0.003208	0.002858
	A_2, B_2	0.129926	0.079593	0.033118	0.017735	0.013484	0.007635	0.003208	0.002857
	A_3, B_3	0.129983	0.079650	0.033121	0.017737	0.013484	0.007636	0.003207	0.002858
	A_4, B_4	0.130007	0.079634	0.033122	0.017737	0.013485	0.007635	0.003207	0.002857
	A_5, B_5	0.129976	0.079635	0.033123	0.017738	0.013485	0.007635	0.003207	0.002858
	A_6, B_6	0.130020	0.079641	0.033123	0.017737	0.013485	0.007635	0.003207	0.002858
	Stand. dev.	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
R-2-D QLCS	A_1, B_1	0.130137	0.079684	0.033115	0.017725	0.013471	0.007626	0.003211	0.002868
	A_2, B_2	0.130812	0.081040	0.034426	0.018990	0.014754	0.009097	0.005383	0.005176
	A_3, B_3	0.130696	0.080593	0.033948	0.018516	0.014272	0.008544	0.004603	0.004380
	A_4, B_4	0.130111	0.080151	0.033605	0.018186	0.013935	0.008164	0.004064	0.003803
	A_5, B_5	0.130010	0.080040	0.033532	0.018148	0.013920	0.008191	0.004128	0.003880
	A_6, B_6	0.130033	0.080041	0.033507	0.018095	0.013847	0.008069	0.003923	0.003655
	Stand. dev.	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000



Fig. 2. Scaled images of Lena for scale invariance tests: (a) 64×64 pixels, (b) 128×128 pixels, (c) 256×256 pixels, (d) 369×369 pixels, (e) 631×631 pixels, (f) 775×775 pixels, (g) 913×913 pixels, and (h) 1024×1024 pixels.

As shown in Tables 5 and 6, compared with the other methods, the standard deviations of different moments of 2-D QLCS under image scaling are almost the smallest. Hence, the experimental results show that the 2-D QLCS has better performance than QPHFM, QPCET and QZM in scale invariance.

4.3. Image reconstruction

In this subsection, several comparison experiments of image reconstruction between L-2-D QLCS, R-2-D QLCS, QZM, QPHFM and quaternion polar complex exponential transform (QPCET) which is one form of QPHTs are presented to evaluate the image reconstruction



Fig. 3. Scaled images of pepper for scale invariance tests: (a) 64×64 pixels, (b) 128×128 pixels, (c) 256×256 pixels, (d) 369×369 pixels, (e) 631×631 pixels, (f) 775×775 pixels, (g) 913×913 pixels, and (h) 1024×1024 pixels.

Table 3

Left- and right-side 2-D QLCS invariants of the same color image Lena under scaling.

		$M_{0,0}^q$	$M_{1,0}^q$	$M_{1,1}^q$	$M_{2,1}^q$	$M_{1,2}^q$	$M_{3,1}^q$	$M_{1,3}^q$	$M_{2,2}^q$
L-2-D QLCS	Original image	1.38100	0.26222	0.22611	0.06097	0.05109	0.04365	0.07298	0.10343
	64×64	1.38078	0.26106	0.22885	0.06252	0.05444	0.04697	0.07472	0.10286
	128×128	1.38108	0.26186	0.22811	0.06150	0.05249	0.04487	0.07383	0.10398
	256×256	1.38079	0.26219	0.22669	0.06114	0.05150	0.04401	0.07318	0.10357
	369×369	1.38056	0.26232	0.22631	0.06097	0.05120	0.04376	0.07303	0.10366
	631×631	1.38079	0.26226	0.22583	0.06092	0.05098	0.04360	0.07291	0.10342
	775×775	1.38072	0.26230	0.22576	0.06087	0.05092	0.04350	0.07284	0.10337
	913×913	1.38081	0.26227	0.22565	0.06088	0.05089	0.04348	0.07283	0.10327
	1024×1024	1.38084	0.26230	0.22560	0.06085	0.05086	0.04346	0.07281	0.10330
	Stand. dev.	0.00015	0.00041	0.00117	0.00054	0.00118	0.00115	0.00064	0.00030
R-2-D QLCS	Original image	1.37672	0.25646	0.24332	0.06251	0.03586	0.04343	0.07683	0.10651
	64×64	1.37644	0.25581	0.24557	0.06404	0.03910	0.04719	0.07968	0.10586
	128×128	1.37677	0.25633	0.24511	0.06304	0.03717	0.04481	0.07810	0.10708
	256×256	1.37650	0.25650	0.24379	0.06267	0.03622	0.04386	0.07719	0.10664
	369×369	1.37628	0.25659	0.24347	0.06252	0.03593	0.04356	0.07695	0.10674
	631×631	1.37651	0.25649	0.24304	0.06246	0.03574	0.04337	0.07674	0.10649
	775×775	1.37645	0.25652	0.24298	0.06241	0.03569	0.04327	0.07665	0.10644
	913×913	1.37654	0.25648	0.24288	0.06242	0.03567	0.04324	0.07663	0.10634
	1024×1024	1.37657	0.25651	0.24284	0.06240	0.03563	0.04322	0.07659	0.10637
	Stand. dev.	0.00015	0.00023	0.00100	0.00053	0.00114	0.00130	0.00102	0.00033

Table 4

Left- and right-side 2-D QLCS invariants of the same color image pepper under scaling.

		$M_{0,0}^q$	$M_{1,0}^q$	$M_{1,1}^q$	$M_{2,1}^q$	$M_{1,2}^q$	$M_{3,1}^q$	$M_{1,3}^q$	$M_{2,2}^q$
L-2-D QLCS	Original image	1.14680	0.23646	0.12919	0.12964	0.05745	0.06625	0.03330	0.09021
	64×64	1.14384	0.23645	0.12629	0.12762	0.05502	0.06738	0.03393	0.08153
	128×128	1.14574	0.23677	0.12815	0.12865	0.05659	0.06656	0.03361	0.08709
	256×256	1.14621	0.23667	0.12874	0.12928	0.05718	0.06630	0.03341	0.08910
	369×369	1.14635	0.23665	0.12897	0.12946	0.05738	0.06622	0.03341	0.08992
	631×631	1.14678	0.23651	0.12916	0.12969	0.05751	0.06620	0.03331	0.09039
	775×775	1.14682	0.23648	0.12924	0.12974	0.05757	0.06613	0.03329	0.09052
	913×913	1.14690	0.23643	0.12925	0.12978	0.05761	0.06614	0.03324	0.09057
	1024×1024	1.14698	0.23645	0.12927	0.12980	0.05763	0.06611	0.03324	0.09065
	Stand. dev.	0.00099	0.00012	0.00097	0.00073	0.00084	0.00040	0.00022	0.00298
R-2-D QLCS	Original image	1.15739	0.26610	0.11874	0.10974	0.04825	0.05599	0.04535	0.07316
	64×64	1.15510	0.26402	0.11733	0.10627	0.04654	0.05705	0.04651	0.06425
	128×128	1.15667	0.26553	0.11831	0.10832	0.04763	0.05626	0.04585	0.07001
	256×256	1.15692	0.26602	0.11850	0.10926	0.04806	0.05602	0.04557	0.07209
	369×369	1.15699	0.26620	0.11860	0.10956	0.04816	0.05596	0.04548	0.07289
	631×631	1.15735	0.26620	0.11871	0.10983	0.04827	0.05595	0.04536	0.07335
	775×775	1.15738	0.26623	0.11877	0.10990	0.04828	0.05590	0.04538	0.07347
	913×913	1.15745	0.26619	0.11876	0.10994	0.04830	0.05591	0.04536	0.07352
	1024×1024	1.15752	0.26623	0.11879	0.10997	0.04830	0.05590	0.04536	0.07360
	Stand. dev.	0.00076	0.00072	0.00047	0.00122	0.00058	0.00037	0.00039	0.00306

performance of the 2-D QLCS. Based on Eqs. (16) and (17), the image can be reconstructed approximately by finite 2-D QLCS. Let K be a constant, the number of moments used in image reconstruction can be

limited as QZM: $n - |m| = \text{even}$, $|m| \leq n \leq K$; QPHFM: $n + |m| \leq K$; QPCET, L-2-D QLCS, R-2-D QLCS: $|n| + |m| \leq K$.

Experiment 1. Figs. 4 and 5 shows some reconstructed images for Lena and mandrill in Fig. 1 (moment order $K_{max} = 5, 25, 45, 50$,

Table 5

Comparison of the standard deviations of L-2-D QLCS, R-2-D QLCS, QPHFM, QPCET and QZM under image scaling for color images Lena. Note that, * indicates that there is no value.

	$M_{0,0}^q$	$M_{2,0}^q$	$M_{2,2}^q$	$M_{2,-2}^q$	$M_{2,-1}^q$	$M_{1,-4}^q$	$M_{1,4}^q$	$M_{2,-3}^q$	$M_{2,-3}^q$
L-2-D QLCS	0.00015	0.00195	0.00030	0.00113	0.00027	0.00041	0.00065	0.00034	0.00034
R-2-D QLCS	0.00015	0.00202	0.00033	0.00166	0.00026	0.00024	0.00067	0.00031	0.00036
QPHFM	0.00382	0.00659	0.00284	0.00289	0.00039	0.01391	0.01342	0.00208	0.00105
QPCET	0.00013	0.00183	0.00196	0.00137	0.00038	0.00098	0.00074	0.00056	0.00051
QZM	0.00013	0.00164	0.00172	0.00177	*	*	*	*	*

Table 6

Comparison of the standard deviations of L-2-D QLCS, R-2-D QLCS, QPHFM, QPCET and QZM under image scaling for color images pepper. Note that, * indicates that there is no value.

	$M_{0,0}^q$	$M_{2,0}^q$	$M_{2,2}^q$	$M_{2,-2}^q$	$M_{2,-1}^q$	$M_{1,-4}^q$	$M_{1,4}^q$	$M_{2,-3}^q$	$M_{2,-3}^q$
L-2-D QLCS	0.00099	0.00140	0.00298	0.00079	0.00091	0.00046	0.00056	0.00026	0.00055
R-2-D QLCS	0.00076	0.00111	0.00306	0.00084	0.00064	0.00044	0.00052	0.00028	0.00067
QPHFM	0.00046	0.00183	0.00535	0.00539	0.00067	0.00411	0.00465	0.00173	0.00179
QPCET	0.00067	0.00039	0.00113	0.00082	0.00100	0.00137	0.00068	0.00122	0.00017
QZM	0.00067	0.00469	0.00133	0.00152	*	*	*	*	*

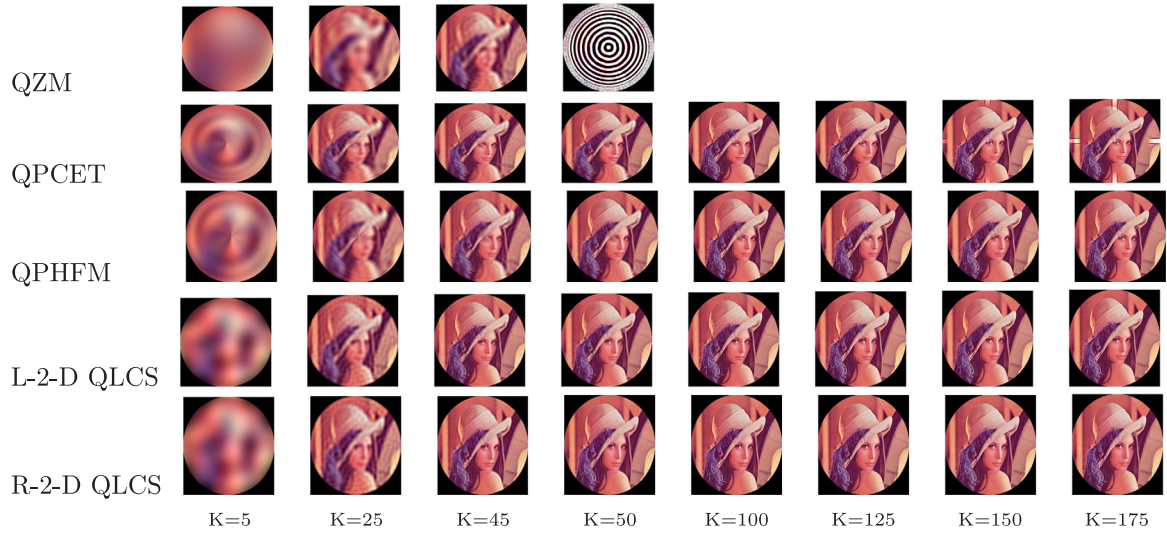


Fig. 4. Reconstructed images for Lena with 512×512 pixels (moment order $K = 5, 25, 45, 50, 100, 125, 150, 175$). (a) QZM, (b) QPCET, (c) QPHFM, (d) L-2-D QLCS, (e) R-2-D QLCS.

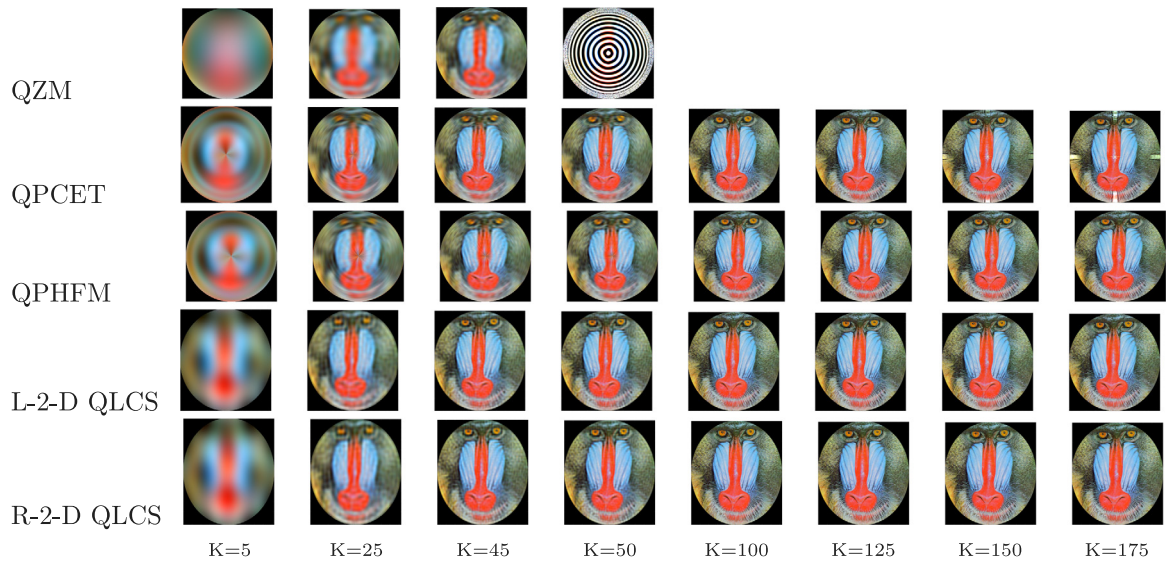


Fig. 5. Reconstructed images for mandrill with 512×512 pixels (moment order $K = 5, 25, 45, 50, 100, 125, 150, 175$). (a) QZM, (b) QPCET, (c) L-2-D QLCS, (d) R-2-D QLCS.

100, 125, 150, 175). Fig. 6 shows the plots of RMSEs for 4 color images in Fig. 1. Seen from Figs. 4–6, the L-2-D QLCS and R-2-D QLCS proposed

in this paper perform much better than QZM, QPCET and QPHFM in image reconstruction. We can also observe that the reconstructed

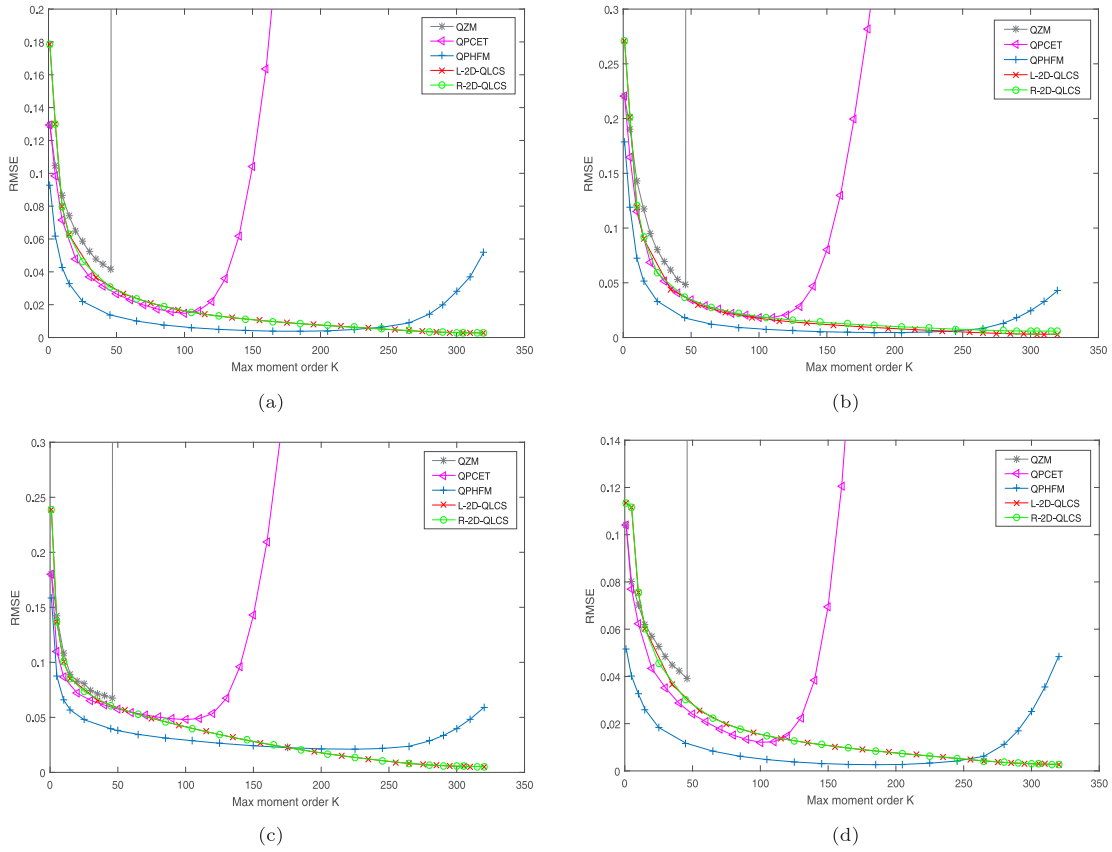


Fig. 6. Comparison of RMSEs of QZM, QPCET, QPHFM, L-2-D QLCS and R-2-D QLCS for color images with 512×512 pixels: (a) Lena, (b) pepper, (c) mandrill, and (d) plane.



Fig. 7. The noisy images of Fig. (a) are added with Gaussian noise. From left to right the variance is: 0.05, 0.1, 0.15, 0.2.

images get closer to the original images with more moments added to the reconstruction process. For QPHFM, the performance of image reconstruction is degraded when the max moment order K exceeds 190. Similar phenomenon happens to QZM and QPCET, but L-2-D QLCS and R-2-D QLCS. Therefore, the numerical stabilities of L-2-D QLCS and R-2-D QLCS are better than QZM, QPCET and QPHFM.

Experiment 2. The quality of the reconstructed image may be seriously influenced by noise [28]. In order to test the image reconstruction performance of 2-D QLCS under noise, the image Lena given in Fig. 1(a) is perturbed by Gaussian noise with variance 0.05, 0.1, 0.15, 0.2, respectively. See Fig. 7.

Fig. 8 shows the plots of RMSEs for 4 color images in Fig. 7. It can be seen that the plots of RMSEs perturbed by Gaussian noise have a similar trend to Fig. 6(a). All our experimental results in this subsection show that the proposed 2-D QLCS has better performance than the generally used QZM, QPCET and QPHFM in numerical stabilities and image representation under noise-free and noisy conditions.

5. Discussion

We have proved that the 2-D QLCS is more suitable for color image processing than 2-D LCTS, and has better performance than the generally used QZM, QPCET and QPHFM in scale invariance, numerical stabilities and image representation. The 2-D QLCS is defined

in Cartesian coordinates, without the instability of calculation results caused by polar coordinate transformation. As show in Fig. 6(a), for QPHFM, the performance of image reconstruction is degraded when the max moment order K exceeds 190. Similar phenomenon happens to QZM and QPCET, but L-2-D QLCS and R-2-D QLCS. Therefore, the numerical stabilities of L-2-D QLCS and R-2-D QLCS are better than QZM, QPCET and QPHFM. The 2-D QLCS has six free parameters, and it is therefore worthy and interesting to investigate the sensitivity of parameters. In this paper, we have not analyze the parameter sensitivity by consistence of the decomposing process and the reconstructing process. This interesting problem that will be studied in the future research.

6. Conclusions

In this paper, motivated by the 2-D QLCT and based on the quaternion theory, we proposed the novel two-dimensional quaternion linear canonical series (2-D QLCS) for color image analysis. The capabilities of the proposed 2-D QLCS on image representation and reconstruction are analyzed, and the relationships between 2-D QLCS and 2-D LCT are investigated in detail. Finally, several comparison experiments are carried out to test the performance of the proposed 2-D QLCS. Experimental results verify the correctness of the proposed method, and show that the proposed 2-D QLCS has better performance than the generally used

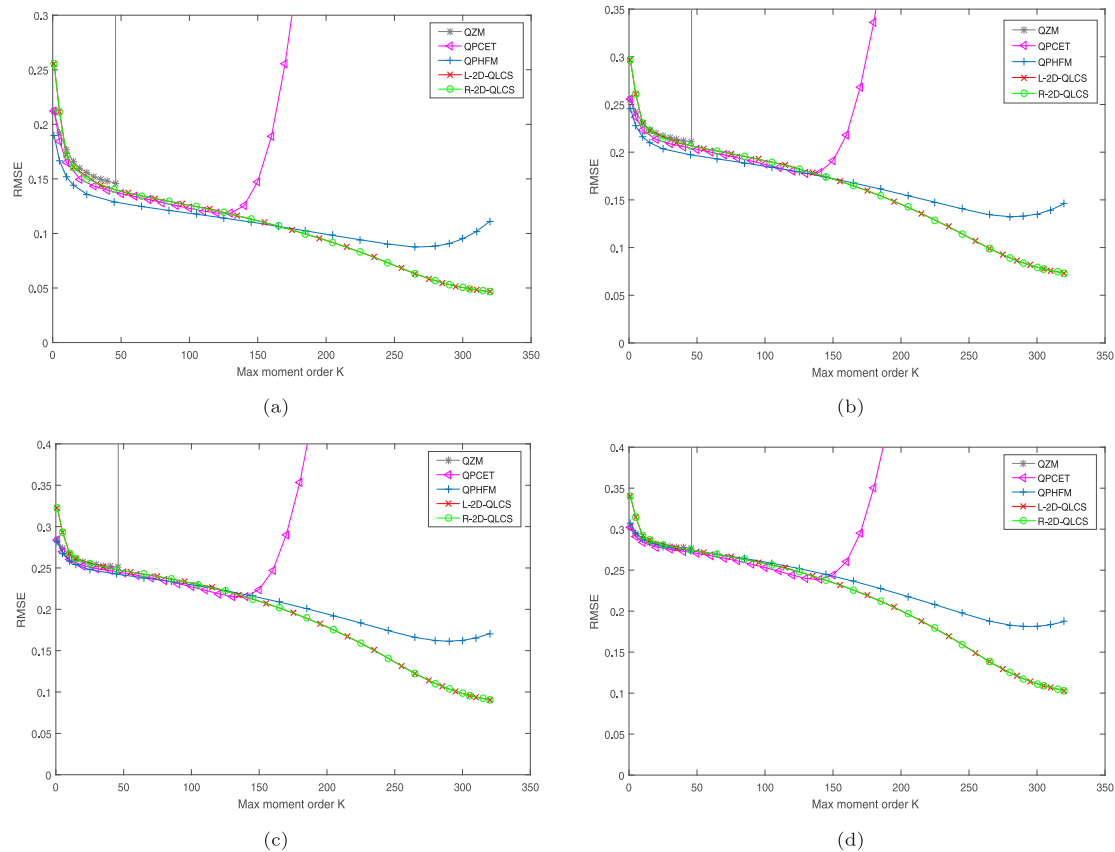


Fig. 8. Comparison of RMSEs of QZM, QPCET, L-2-D QLCS and R-2-D QLCS for color images Lena given in Fig. 1(a) with Gaussian noise of variance: (a) variance 0.05, (b) variance 0.1, (c) variance 0.15, and (d) variance 0.2.

QZM, QPCET and QPHFM in scale invariance, numerical stabilities and image representation. In future work, we will apply the proposed 2-D QLCS in a variety of applications, such as color image watermarking, color image segmentation, color image pattern recognition, and so on.

CRedit authorship contribution statement

Zhen-Wei Li: Modeling, Algorithm programming, Experiments, Original draft preparation. **Bing-Zhao Li:** Supervision, Presentation reviewing. **Min Qi:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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